

PRACTICE EXAM 1 OF 4 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 1

Twenty questions, mixed domains, weighted to the real paper. Nine seconds per question.

QUESTIONS 20 DOMAINS all five PAUSE 9 seconds

How to sit this

- ◆ **Answer all twenty before checking any.** Checking as you go inflates your score and teaches nothing.
- ◆ **Write the letter down.** A remembered answer becomes the right one by the time the walkthrough finishes.
- ◆ **Note which domain each miss came from.** The scoring slide at the end needs it.
- ◆ **Do not look anything up.** A score you helped is not a measurement.

ON THE EXAM

Twenty questions is a snapshot, not a verdict. The pattern across four papers is the useful thing.

A logistics company runs a shipment-status service. Every request performs the same three operations in the same order: validate the tracking number, fetch the carrier record, and format a customer-facing summary. Each step is independent of the others and the order never varies.

The current implementation gives a single agent all three tasks and lets it decide how to proceed. Output is inconsistent, and about one response in twelve omits the carrier detail entirely.

What is the most effective design change?

- ☐ A Add emphasis to the system prompt requiring all three steps every time
- ☐ B Restructure as prompt chaining, with three sequential calls each handling one step
- ☐ C Give the agent three tools, one per step, and let it call them
- ☐ D Increase max_turns so the agent has room to complete all three steps

◆ PAUSE AND DECIDE

What is the most effective design change?

- A

 Emphasis in the prompt
Reduces the omission rate without eliminating it. Symptom, not cause.
- B

 Prompt chaining
Same steps, same order, independent. There is no decision for an agent to make.
- C

 Three tools
Better than one blob, and the agent still owns the path, so it can still skip a step.
- D

 Raise max_turns
Nothing suggests it ran out of turns. It chose not to do the work.

THE SIGNAL

Could you draw the flowchart in advance? Yes, so it is a workflow. Module 2.

A regulated organisation must ensure no Claude Code session anywhere in the company can execute curl. Engineering teams control their own repositories and their own settings files.

A platform engineer adds a deny rule for Bash curl to every project's `.claude/settings.json` and enforces it through code review.

What is the weakness in this approach?

- ☐ A Deny rules cannot match command arguments, only tool names
- ☐ B A project setting can be overridden by a local setting, and a repository can be missed
- ☐ C Code review cannot detect settings file changes
- ☐ D There is no weakness; project scope is the correct layer for a team-wide rule

◆ PAUSE AND DECIDE

What is the weakness in this approach?

- A

 Cannot match arguments
Permissions can match command patterns. This is not the gap.
- C

 Review cannot detect it
Review can see a settings diff. Review just cannot see the repository nobody created a rule in.
- B

 Local overrides, and missed repos
settings.local.json sits above project scope, and every repository is a place to forget.
- D

 No weakness
"Anywhere in the company" needs the managed layer, which nothing below can relax.

THE SIGNAL

Enterprise managed policy is the only layer nothing below can override. Module 8.

A single prompt asks Claude to extract every regulatory citation from a filing and, for each one, assess whether the filing complies with it.

When compliance assessments are requested, several citations go missing. When assessments are omitted, citation recall is high. The assessments themselves are consistently well reasoned.

Both outputs are required.

What is the most effective approach?

- ☐ A Instruct the model that no citation may be missed, regardless of assessment effort
- ☐ B Two calls: extract citations, then assess the extracted list
- ☐ C Ask for citations first, then assessments, within the same response
- ☐ D Increase max_tokens so there is room for both outputs

◆ PAUSE AND DECIDE

What is the most effective approach?

- A

 Instruct completeness
Swaps which half suffers. Emphasis redistributes the trade rather than removing it.
- C

 Sequence in one response
A partial fix. Both jobs still share one attention budget.
- B

 Two calls
Recall gets a call to itself, then assessment gets a call to itself on a fixed list.
- D

 More output tokens
Nothing was truncated. The assessments were complete and well reasoned.

THE SIGNAL

One prompt, two jobs, both worse. Module 12.

An agent has been running a support session for ninety minutes. At the start, the user stated that their organisation cannot accept third-party cookies for legal reasons.

The agent now recommends a solution requiring third-party cookies. The constraint is still present in the conversation history, nothing was truncated, and no error was raised.

What is the most likely explanation?

- ☐ A The context window overflowed and silently dropped early turns
- ☐ B Attention degrades as the window fills, so early information receives less weight
- ☐ C The constraint was stated by the user rather than in the system prompt, so it is not binding
- ☐ D Compaction fired and removed the constraint from the summary

◆ PAUSE AND DECIDE

What is the most likely explanation?

- A

 Overflow
Overflow raises an error. The stem says nothing was truncated and nothing errored.
- B

 Attention degradation
Present but under-weighted. Silent, gradual, and worse as the session lengthens.
- C

 Not binding from a user
Invented rule. It applied the constraint correctly for most of ninety minutes.
- D

 Compaction removed it
The stem says the constraint is still present in the history.

THE SIGNAL

Still present, nothing truncated, no error. That trio is the degradation signature. Module 15.

A coordinator spawns five subagents to research different aspects of a competitor analysis. Each is briefed with one line: "Research this area and report your findings."

Two problems recur. Three subagents independently researched the same pricing history. And a market the client specifically named appears in no subagent's output.

What is the most effective fix?

- ☐ A Add a sixth subagent covering the missing market
- ☐ B Give each subagent a narrow objective, explicit out-of-scope boundaries, and a required output format
- ☐ C Have the coordinator deduplicate overlapping sections during synthesis
- ☐ D Run the subagents sequentially so each can see what the previous one covered

◆ PAUSE AND DECIDE

What is the most effective fix?

- A

 A sixth subagent
Patches this gap and not the next. The briefs still allow overlap and holes.
- C

 Deduplicate at synthesis
Hides the wasted work and does nothing about the missing market.
- B

 Narrow objectives and boundaries
Overlap and gaps are one defect: nobody was told precisely what was theirs.
- D

 Run sequentially
Subagents cannot see each other's work in any order. Slower, same problems.

THE SIGNAL

Duplicated coverage and missing coverage are one problem with two faces. Module 3.

A support agent has a tool described as: "Checks entitlement." Its input schema takes a `customer_id` and a `product_code`.

In production the agent never calls it. It answers entitlement questions from the conversation and from a general knowledge tool, and is frequently wrong.

The tool works correctly when called manually.

What is the most likely cause?

- ☐ A The tool is missing from the agent's allowed tools list
- ☐ B The description does not describe a situation the agent recognises it is in
- ☐ C The input schema requires two parameters, which the agent cannot supply
- ☐ D The general knowledge tool is ranked higher in the tool list

◆ PAUSE AND DECIDE

What is the most likely cause?

- A

 Not in allowedTools
Possible, and then it would not appear at all. The stem implies it is available.
- B

 Description does not match a situation
"Checks entitlement" says nothing about when to use it or what it returns.
- C

 Two required parameters
Both are things a support agent routinely has. Not the obstacle.
- D

 Ranking in the tool list
Invented. There is no ranking that decides selection.

THE SIGNAL

Never uses the tool at all usually means the description does not describe a recognisable situation. Module 5.

A media monitoring service classifies 300,000 articles a month for sentiment. Two consumers use the output: a live alerting dashboard needing results within a minute, and a monthly trend report.

About 6% of articles match alerting keywords. The rest feed only the monthly report.

All 300,000 currently go through synchronous requests and cost is the concern.

What is the most cost-effective change?

- ☐ A Move all articles to the Batch API
- ☐ B Keep the 6% synchronous and route the remaining 94% through the Batch API
- ☐ C Process all articles in parallel to reduce runtime
- ☐ D Use a smaller model for all classifications

◆ PAUSE AND DECIDE

What is the most cost-effective change?

- A

 Batch everything
The alerting dashboard needs a minute. Batch offers up to 24 hours.
- C

 Parallel
Buys wall-clock time. Costs exactly the same.
- B

 Route by urgency
94% at half price, alerting path untouched. A keyword rule already sorts them.
- D

 Smaller model everywhere
Trades accuracy on the alerting path, which is where it matters most.

THE SIGNAL

Batch-or-sync is often a false binary. Route by urgency. Module 14.

An overnight agent reconciles supplier invoices. It is invoked with a turn cap and a budget cap. The orchestration wrapper marks a run complete whenever the invocation returns without throwing.

An auditor finds reconciliations marked complete where a quarter of the invoices were never examined.

What is the root cause?

- ☐ A The turn cap is set too low for the invoice volume
- ☐ B The wrapper treats any non-exception return as success, including runs aborted by a cap
- ☐ C The agent should report when it has not finished
- ☐ D The work should be split across several smaller invocations

◆ PAUSE AND DECIDE

What is the root cause?

- A

 Cap too low
Contributing. Raise it and the same silent failure returns at a larger volume.
- B

 Non-exception treated as success
A cap firing is a normal return. The wrapper cannot tell finished from stopped.
- C

 Agent should report
A run killed by a cap has no turn left in which to report anything.
- D

 Smaller runs
Each smaller run still reports a capped abort as a success.

THE SIGNAL

Hitting a guardrail is neither an error nor a completion. It is a third outcome. Module 4.

A team requires that Claude Code never runs a deployment command against the staging environment during the nightly data load, which runs from 01:00 to 04:00.

Outside that window, staging deployments are routine and frequent.

Deployment commands are issued through Bash.

What enforces this?

- ☐ A A permissions deny rule on Bash deployment commands
- ☐ B A PreToolUse hook on Bash that checks the current time and blocks within the window
- ☐ C A CLAUDE.md rule stating the nightly load window
- ☐ D A PostToolUse hook that alerts when a deployment runs during the window

◆ PAUSE AND DECIDE

What enforces this?

- A

 Permissions deny
Enforcement, and permanent. Staging deploys are routine outside the window.
- C

 CLAUDE.md rule
Context. It persuades and cannot guarantee.
- B

 PreToolUse with a time check
Conditional enforcement. It evaluates something a permission cannot express.
- D

 PostToolUse alert
The deployment already ran during the data load.

THE SIGNAL

Permissions are on or off. Hooks can evaluate a condition. Module 9.

An MCP server exposes a currency conversion tool. When an unsupported currency code is passed, the server raises an exception that the SDK converts into a JSON-RPC error.

Traces show the agent calling the tool with the same unsupported code up to four times, then telling the user conversion is unavailable.

What should change?

- ☐ A Report the unsupported currency as a tool execution result with `isError` and a message naming the supported codes
- ☐ B Reduce the retry count so the agent gives up sooner
- ☐ C Have the tool default to USD when the code is unsupported
- ☐ D Add the supported currency list to the system prompt

◆ PAUSE AND DECIDE

What should change?

- A

 isError with the supported codes
The model reads the message and can correct the call. Protocol errors are invisible to it.
- B

 Fewer retries
Four identical attempts. A second identical attempt is no better than a fourth.
- C

 Default to USD
Silently returns a wrong conversion. A loud failure becomes a quiet one.
- D

 List in the system prompt
Helps a little, and the next unsupported input still fails invisibly.

THE SIGNAL

The same unsupported code four times means the agent learned nothing. Module 6.

A migration agent works across three sessions over two weeks. Step nine needs a schema decision made in session one and a list of affected services discovered in session two.

Compaction fires several times within each session. Everything currently lives in the conversation.

What should they implement?

- ☐ A Raise the compaction threshold so early turns survive longer
- ☐ B Write decisions and findings to a state file as each step completes, and read from it later
- ☐ C Repeat the session-one decision at the start of every prompt
- ☐ D Run each step as a subagent so context stays isolated

◆ PAUSE AND DECIDE

What should they implement?

- A

 Higher threshold
Delays compaction within a session. Does nothing across three sessions.
- B

 A state file
Survives compaction and survives the session. Both requirements, one mechanism.
- C

 Repeat it every prompt
Works for one known decision. Findings you cannot predict are not covered.
- D

 Subagents per step
Subagents inherit nothing, so step nine would see even less.

THE SIGNAL

Two requirements: survive compaction, and survive the session. Only external state does both. Module 15.

A coordinator has three genuinely independent investigations to run. `allowedTools` includes the delegation tool, and the system prompt instructs it to work in parallel wherever possible. Traces show one delegation call, then a wait for its result, then the next. Total time is roughly three times the estimate. Each subagent completes correctly.

What change makes them run concurrently?

- ☐ A Strengthen the system prompt to require parallel execution
- ☐ B Emit all three delegation calls in a single assistant response
- ☐ C Raise `max_turns` so there is room to issue all three calls
- ☐ D Merge the three investigations into one subagent

◆ PAUSE AND DECIDE

What change makes them run concurrently?

- A

 Stronger wording
The instruction already exists and is not working. Emphasis does not change response shape.
- B

 Three calls in one response
Concurrency is a property of the message, not of the prompt.
- C

 Raise max_turns
It issued all three calls. Turns were never the constraint.
- D

 Merge into one
Removes the parallelism problem by removing the parallelism.

THE SIGNAL

"The system prompt already instructs it" eliminates every option that writes a better instruction. Module 3.

A clinical records extraction schema requires `patient_nhs_number`, `admission_date`, `discharge_date` and `referring_gp_code`, all mandatory.

About 20% of records are for self-referred patients with no referring GP. For those, extraction returns a well-formed code that matches no GP in the register.

The prompt instructs the model not to invent values.

What is the most effective fix?

- ☐ A Add examples showing self-referred records with no GP code
- ☐ B Make `referring_gp_code` nullable, documenting that null means self-referred
- ☐ C Validate returned codes against the GP register and reject non-matches
- ☐ D Lower the temperature to reduce invention

◆ PAUSE AND DECIDE

What is the most effective fix?

- A

 Examples
The prompt already says do not invent. The schema says produce a value. The schema wins.
- B

 Make it nullable
Removes the conflict. Absence becomes expressible, so there is nothing to invent.
- C

 Validate against the register
Catches it afterwards, on every record, forever. The pressure to invent remains.
- D

 Lower temperature
More consistent inventions, not fewer. It still has to fill the field.

THE SIGNAL

Required plus unknowable equals fabricated. "The prompt already instructs" is the eliminator. Module 13.

An engineer needs to rename a single constant across one file and update the comment above it. They know exactly which file and exactly what the change is.

Their team has set plan mode as the project default, so this produces a plan they must read and approve.

What should they do?

- ☐ A Keep the default. Planning is always safer than not planning
- ☐ B Remove the project default and enter plan mode deliberately when a change warrants it
- ☐ C Keep the default but ask for shorter plans on small changes
- ☐ D Remove plan mode entirely and rely on reviewing diffs

◆ PAUSE AND DECIDE

What should they do?

- A

 Always plan
Anthropic's guidance is explicit: if you could describe the diff in one sentence, skip the plan.
- B

 Deliberate, not default
Keeps planning where it pays and removes it where it is pure overhead.
- C

 Shorter plans
Still a plan and still an approval step. It shrinks the friction rather than removing it.
- D

 Remove plan mode
Overcorrects. Multi-file work in unfamiliar code genuinely needs it.

THE SIGNAL

One file, one obvious edit, approach not in question. If you could describe the diff in one sentence, skip the plan. Module 10.

A developer is deep in a data model redesign with an agent. They want to try a fundamentally different normalisation approach without losing the current thread, and expect to compare both before choosing.

The agent's context holds substantial reasoning about the current design that the experiment will depend on.

What should they do?

- ☐ A Spawn a subagent to explore the alternative approach
- ☐ B Fork the session, run the experiment in the fork, and compare
- ☐ C Start a fresh session for the alternative approach
- ☐ D Continue in the same session and revert if the experiment fails

◆ PAUSE AND DECIDE

What should they do?

- A

 A subagent
Subagents inherit nothing. The experiment depends on the reasoning the parent holds.
- C

 Fresh session
Discards the reasoning the experiment depends on.
- B

 Fork
Branches the decision, keeps the parent's context, and both branches stay resumable.
- D

 Continue and revert
Reverting files does not revert the conversation. The abandoned approach stays in context.

THE SIGNAL

Fork branches a decision. A subagent isolates context. The stem says the experiment needs the parent's context. Module 16.

A platform team has built an MCP server for their internal feature-flag system. Every engineer working in the payments repository should have it configured automatically on clone, with no setup step.

The same team also uses a personal task-tracking MCP server that should follow each individual across all their repositories.

Which scopes?

- ☐ A Feature flags at user scope; task tracking at project scope
- ☐ B Feature flags at project scope; task tracking at user scope
- ☐ C Both at project scope, so the configuration is version controlled
- ☐ D Both at local scope, so each engineer controls their own setup

◆ PAUSE AND DECIDE

Which scopes?

- A

 User, then project
Reversed. User scope is per person, so every engineer would add the flag server by hand.
- B

 Project, then user
Project ships with the repository. User follows the individual across projects.
- C

 Both project
The team inherits somebody's task tracker, and it only works in that repository.
- D

 Both local
Local is one person, one project. Neither requirement is met.

THE SIGNAL

"Everyone on this repository" is project. "This person everywhere" is user. Module 7.

An agent answers questions about a monorepo of roughly 6,000 files. Its current approach reads every file in the relevant package before answering.

Packages range from 20 to 900 files. Answers on small packages are good; on large ones they degrade and slow noticeably.

Most questions concern two or three files.

What is the most effective change?

- ☐ A Read packages in batches, summarising each batch
- ☐ B Grep for the relevant symbols, read only the matching files, then follow imports
- ☐ C Move to a model with a larger context window
- ☐ D Read files in modification-time order, newest first

◆ PAUSE AND DECIDE

What is the most effective change?

- A

 Batch and summarise
Every file still enters context before being summarised. The summaries are tokens on top.
- B

 Search, read, follow
Context grows with relevance rather than package size.
- C

 Bigger window
Buys headroom at the same fill rate. The strategy still scales with package size.
- D

 Newest first
Reorders the reading without reducing it. Recency is not relevance.

THE SIGNAL

Good on small packages, worse on large ones: the volume read is the variable. Module 16.

A CI reviewer is given a specialist persona using the flag that replaces Claude Code's system prompt entirely.

After the change, review quality falls. Claude ignores repository conventions, occasionally attempts file edits during a read-only review, and its file navigation becomes erratic.

The replacement prompt text itself is well written and correct.

What went wrong?

- ☐ A The persona prompt needs more detail about the repository
- ☐ B Replacing the system prompt removed Claude Code's built-in tool guidance and safety context
- ☐ C The model is too small for specialist review work
- ☐ D CLAUDE.md is not loaded in headless mode

◆ PAUSE AND DECIDE

What went wrong?

- A

 More detail
Treats the symptom. They would be rewriting a default prompt they deleted by accident.
- C

 Model too small
Nothing changed about the model. The behaviour changed when the flag did.
- B

 The default was replaced
Three unrelated regressions at once is the signature of a removed default.
- D

 CLAUDE.md not loading
Explains the conventions. Not the erratic navigation or the attempted edits.

THE SIGNAL

Three unrelated symptoms appearing together means a default went missing, not a weak prompt. Use append instead. Module 11.

A pipeline validates extracted records against a schema. On failure it resends the identical request up to three times.

Analysis shows every retry fails identically. The most common failure is an amount field arriving as "approx 4,500" where the schema requires a number.

Retry cost is three times the successful-path cost on those records.

What should change?

- ☐ A Add exponential backoff between retries
- ☐ B Send the validation error back with the retry, naming the field and required type
- ☐ C Loosen the schema to accept the amount as a string
- ☐ D Increase the retry count to five

◆ PAUSE AND DECIDE

What should change?

- A

 Backoff
For transient failures. Identical repeats mean deterministic.
- B

 Send the error back
The model learns what was wrong and corrects that field. The loop converges.
- C

 Loosen the schema
Accepts bad data to make the error go away. A loud failure becomes a quiet one.
- D

 More retries
Five identical failures instead of three, at more cost.

THE SIGNAL

"Every retry fails identically" means deterministic. No retry policy fixes a deterministic failure. Module 13.

A security team investigates alerts. Each investigation starts from an alert, and what is examined next depends entirely on what the previous step found: a suspicious IP leads to connection logs, which may lead to a process tree, or to a user's authentication history. No two investigations follow the same path. An engineer proposes a fixed five-step pipeline covering the most common route.

What should you advise?

- ☐ A Build the fixed pipeline; it covers the majority of cases
- ☐ B An agentic loop with investigation tools and a turn cap, since the path cannot be known in advance
- ☐ C A fixed pipeline with a human escalation step for unusual cases
- ☐ D Parallelise the five steps so all evidence is gathered at once

◆ PAUSE AND DECIDE

What should you advise?

- A

 Fixed pipeline
The stem says no two investigations follow the same path. A fixed path is wrong by construction.
- B

 Agentic loop with a cap
What is examined next depends on what the last step found. Nobody can draw this flowchart.
- C

 Pipeline plus escalation
Escalates the majority, since unusual is the normal case here.
- D

 Parallelise the five steps
They are a chain. Each step needs the previous result, so they cannot run at once.

THE SIGNAL

"Depends entirely on what the previous step found" is the agent signature, and it also rules out parallelism. Module 1.

Scoring exam 1

- ◆ **16 or more.** Comfortable. Move on and space the remaining papers out.
- ◆ **13 to 15.** Around the pass line. Rewatch the domain your misses cluster in, then sit exam 2.
- ◆ **12 or fewer.** Work through the domain modules again before another paper.
- ◆ **Where the misses cluster matters more than the total.** Four misses in one domain is a different problem from four spread out.

ON THE EXAM

Do not convert this to a percentage and compare it to 720. The real exam is scaled, not raw.

Which domain did you miss?

QUESTION NUMBERS	DOMAIN	MODULES TO REWATCH
1, 5, 8, 12, 15, 20	D1 Agentic architecture	1 to 4
6, 10, 16	D2 Tool design and MCP	5 to 7
2, 9, 14, 18	D3 Claude Code	8 to 11
3, 7, 13, 19	D4 Prompt and structured output	12 to 14
4, 11, 17	D5 Context and reliability	15 to 17

ON THE EXAM

Two or more misses in one row is a signal. One miss in a row is noise at this sample size.